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# Geared Motor Noise Diagnosis Worksheet

For maintenance engineers and OEM designers working with spur, helical, worm and planetary DC geared motors

Work through the steps in order and record each reading. Do not dismantle the gearbox until the isolation test has shown where the noise originates. Keep the completed sheet as the baseline for future comparisons.

## Step 1 — Safety and baseline

- [ ] Isolate and lock out the supply before touching the shaft, guards or gearbox; verify zero voltage at the motor terminals.
- [ ] Record nameplate data: voltage, power, motor speed, gear ratio, gear type and specified lubricant.
- [ ] Measure sound level about 1 m from the gearbox at a fixed position, speed and load; record the background level with the machine stopped.
- [ ] Treat readings less than 3 dB above background as unreliable; no background correction is needed when the difference exceeds 10 dB.
- [ ] Record a spectrum with an FFT analyser or spectrum app and note the frequency of the dominant tone.
- [ ] Note whether the noise is new, a gradual rise, or present since commissioning.

## Step 2 — Describe the sound

- [ ] Whine: does the pitch rise and fall with speed? Yes → gear mesh. No → suspect PWM switching.
- [ ] Grinding or gritty rasp → suspect tooth surface damage or contaminated lubricant.
- [ ] Click or knock → count the events per second and compare with each shaft frequency (Step 3).
- [ ] Knock only when the load reverses → suspect backlash, a loose key or a worn coupling.
- [ ] Low rumble with a warm bearing housing → suspect bearing wear or excessive overhung load.
- [ ] Hiss, squeal or chatter at the motor end → inspect brushes, commutator and brake air gap.

## Step 3 — Calculate expected frequencies

- [ ] Shaft frequency of each shaft: $f = N / 60$, with N in rpm.
- [ ] Mesh frequency of each stage: $f = z \times N / 60$, using the teeth and speed of either gear in that mesh.
- [ ] Planetary stage with fixed ring gear: $f = z(\text{ring}) \times N(\text{carrier}) / 60$.
- [ ] Worm stage: $f$ = number of worm starts × $N(\text{worm}) / 60$.
- [ ] Sidebands at mesh frequency ± a shaft frequency identify the shaft carrying an eccentric or damaged gear.
- [ ] A tone that stays at one frequency at every speed is almost always the PWM carrier or its harmonics.

## Step 4 — Isolation test

- [ ] With power isolated, rotate the output by hand: feel for notchiness, tight spots and free play at reversal.
- [ ] Run the geared motor decoupled from the machine at the same speed; if the noise disappears, inspect the coupling and driven machine.
- [ ] Run from a clean DC source such as a battery or linear supply; if a fixed-pitch tone disappears, the drive is the source.
- [ ] Where the construction allows, run the motor without the gearbox; if the noise disappears, the gearbox is the source.

- [ ] Re-apply load in steps and note how level and character change with load, speed and direction.
- [ ] Sweep the speed slowly; a noise that peaks sharply at one speed indicates resonance of the mounting structure.

## Step 5 — Lubrication and mounting checks

- [ ] Check grease or oil level, colour and consistency; a milky appearance indicates water ingress.
- [ ] Wipe a lubricant sample across a magnet: steel particles indicate active tooth or bearing wear.
- [ ] Confirm the lubricant grade matches the gearbox maker's specification; never mix grease thickener types.
- [ ] Check mounting bolt torque and inspect the bracket for cracks; thin sheet-metal brackets amplify mesh tones.
- [ ] Check belt tension and sprocket position against the gearbox overhung load rating.
- [ ] Check alignment between the gearbox output shaft and the driven shaft.

## Step 6 — Corrective actions and sign-off

- [ ] Raise the PWM switching frequency to 16–20 kHz or above, within the driver's thermal and EMC limits.
- [ ] Replace gears showing pitting, chipped teeth or scuffing as matched pairs; replace damaged bearings.
- [ ] Clean out and refill the lubricant to the specified grade and quantity; do not overfill.
- [ ] Stiffen or damp the mounting, or change speed or ratio to move the mesh tone away from a resonance.
- [ ] Re-measure sound level and spectrum after the repair and file the readings as the new baseline.

## Sound signature quick reference

| Sound | Frequency clue | Likely cause | First action |
|---|---|---|---|
| Whine | Tracks speed; at mesh frequency | Gear wear, backlash, misalignment | Spectrum check; inspect teeth and lubricant |
| Fixed-pitch whistle | Constant at every speed | PWM switching in audible band | Run on battery; raise carrier frequency |
| Grinding | Broadband and rough | Tooth damage or contamination | Inspect lubricant for particles |
| Click or knock | Once per shaft revolution | Chipped tooth on that shaft | Count rate; match to shaft frequency |
| Knock at reversal | Only on direction change | Backlash, loose key, coupling | Check free play by hand |
| Rumble | Non-tonal, rises with load | Bearing wear or overhung load | Stethoscope at bearings; check radial load |
| Squeal or scrape | Motor end, at motor speed | Brushes, glazed commutator, brake rub | Inspect brushes and brake air gap |

*Sound levels and frequencies are diagnostic guides, not acceptance limits. Confirm lubricant grades, load ratings and noise limits against the gearbox manufacturer's data for your unit.*

**Need help sizing or replacing a motor?**
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

*General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.*